

BACSWN Master Framework Schedule

BACSWN — Master Framework Schedule & Milestone Plan

Document: BACSWN-SCHED-2026-003 **Classification:** CONFIDENTIAL — For Authorized Recipients Only **Prepared for:** Owner & CEO, BACSWN (private delivery partner) **Prepared by:** Sky Miles Limited — AI Elevate Division **Version:** 1.8 — June 2026 **Delivery model:** Build-Own-Operate (BOO) public-private partnership with the Bahamian government. **Program:** 24-month implementation, 4 phases, 18 parallel workstreams. **Scope:** This is an **implementation plan** — workstreams, milestones, dependencies, and risks. Financial and commercial modeling is intentionally **out of scope** and handled separately. **Change log:** v1.8 adds the consolidated **SWOT & Risk Analysis** (BACSWN-RISK-2026-008) as the authoritative living risk register; §7 here is the schedule-impacting subset. v1.7 adds the **Training & Capacity-Building Plan** (BACSWN-TRN-2026-007) as workstream WS-R with gates TRN-1/TRN-2, and narrows WS-I to operations/O&M/handover. v1.6 adds the **Master V&V and Interface Control (ICD) Plan** (BACSWN-VV-2026-006) — gates VV-1 (V&V plan + ICD register baselined) and VV-2 (operational validation) — and broadens WS-H to cover FAT/SAT/IV&V and ICD management. v1.5 adds the **Environmental, Land & Permitting** workstream (WS-Q) with gates ENV-1/ENV-2 — land rights, environmental impact assessment, protected-area clearance, and building/civil-aviation siting permits — which gate all civil works (risk R12). v1.4 integrates four Day-2 / governance items — **hub disaster-recovery failover (DR-1)**, **O&M / sustainment readiness (OM-1)**, **data-governance & sovereignty sign-off (GOV-1)**, and **lightning/grounding/corrosion in structural design (STR-1)** — and adds the O&M Plan (BACSWN-OM-2026-004) and Data Governance & Sovereignty Policy (BACSWN-GOV-2026-005) as companions. v1.3 removes budget/financial framing to keep this a pure implementation plan. v1.2 added the **primary surveillance radar program** (Raytheon radar at designated radar towers) as workstream WS-P with gates RAD-1...RAD-4, long-lead defense procurement, **US export/ITAR licensing**, and heavy radar-tower civil works; and wired in the **Tomorrow.io** space-based-radar weather feed (WS-F). v1.1 added workstreams/gates for cryptographic security, Cat 4/5 structural certification, emergency-services integration, commercial+military surveillance & overflight-fee metering, the IoT sensor network, drones/UAS, and external/U.S. interoperability.

1. How to Read This Schedule

- Timeline:** 24 months, Month 0 (M0) = program kickoff. Anchored to a **south-to-north** hardware rollout so the highest hurricane-exposure stations are protected before each June 1.
- Workstreams** run in parallel; **milestones (M-gates)** are go/no-go decision points with defined exit criteria. A phase does not advance until its gate is signed off by the program's technical authority (owner/CEO).
- Critical path** is marked ☒. Hurricane-season constraints are marked Ⓢ.

2. Phase Overview

Phase	Window	Theme	Headline outcome
Phase 0	M0 (pre-kickoff)	Mobilization & authorizations	Governance, URCA filing, Raytheon/ITAR procurement underway
Phase 1	M1–M6	Foundation	4 pilot stations live on VHF + Iridium; radar program kicked off
Phase 2	M7–M12	Intelligence	Edge AI on pilots; consensus & autonomy proven; security hardened
Phase 3	M13–M18	Full Network	All 15 stations; emergency-services integration; primary radar + surveillance live
Phase 4	M19–M24	Advanced Capabilities	Propagation forecasting, IoT/drones, full autonomy, handover

Scope note. Beyond the core mesh, this plan sequences additive capability workstreams — primary surveillance radar (Raytheon, WS-P), emergency-services integration, commercial+military surveillance & fee metering, the IoT sensor network, drones/UAS, and external/U.S. interoperability. WS-P (Raytheon radar) is the largest long-lead item and is started at M1. Fee metering (REV-1) is built early so it is operational as soon as the airspace is monitored. *(Commercial/financial modeling is out of scope for this implementation plan.)*

3. Workstreams

ID	Workstream	Owner	Spans
WS-A	Program & Governance (PMO, owner/CEO reporting, change control)	Sky Miles PMO	M0–M24

ID	Workstream	Owner	Spans
WS-B	Regulatory & Spectrum (URCA licensing, BCAA/ICAO coordination) ☒☒	Regulatory lead	M0–M9
WS-C	Comms Network Engineering (VHF mesh, satellite, cellular, link budgets)	Network eng	M1–M18
WS-D	Edge Node Hardware (Jetson, radios, sensors, power, enclosures)	Hardware eng	M1–M18
WS-E	Software & Mesh Protocol (stack, routing, store-and-forward, OTA)	Platform eng	M1–M22
WS-F	Edge AI & Forecast Verification (microclimate models, MOS, consensus)	AI/ML	M5–M22
WS-G	Site Civil & Power (towers, solar, battery, grid tie, enclosures)	Civil/field	M2–M18
WS-H	Integration, V&V & Acceptance (FAT/SAT, IV&V, ICD management, RTM, commissioning, drills)	QA / V&V	M2–M24
WS-I	Operations, O&M & Handover (NOC, watch officers, field service, spares, runbooks)	Ops	M10–M24
WS-J	Security & Cryptography (PKI, secure boot/HSM, mTLS, SecOps, pen-test)	Security	M1–M24
WS-K	Emergency-Services Integration (CAP/EDXL, COP, shelter & evacuation, RBDF)	Integration	M8–M22
WS-L	Airspace Surveillance & Fee Accounting (commercial+military fusion, overflight/landing-fee metering, CORSIA)	Surveillance	M4–M20
WS-M	IoT Sensor Network (LoRaWAN/NB-IoT gateways, surge/flood/lightning sensors)	IoT eng	M10–M22
WS-N	Drone / UAS Program (survey, recon, SAR, comms relay, deconfliction)	UAS lead	M16–M24
WS-O	External / International Interoperability (ICAO OPMET/WAFS/AFTN, U.S./NORAD) — phased	Integration	M18–M24+
WS-P	Primary Surveillance Radar — Raytheon (procurement, export/ITAR, radar-tower civil works, install, calibration, integration) ☒☒	Radar / Civil	M1–M20
WS-Q	Environmental, Land & Permitting (land rights/leases, EIA, protected-area clearance, building & civil-aviation siting permits) ☒☒	Legal / Env	M0–M12
WS-R	Training & Capacity Building (role-based curricula, certification, train-the-trainer, Bahamian workforce, academic partnerships)	Training	M6–M24

Cat 4/5 **structural survivability certification** is delivered within WS-G (Site Civil & Power) and gated explicitly (STR-1, plus per-station sign-off at acceptance). IoT (WS-M) uses license-exempt ISM spectrum, so it does **not** depend on the URCA VHF licensing critical path.

4. Milestone Register

4.1 Core Network Gates

Gate	Month	Milestone	Exit criteria (go/no-go)	Critical
M0	0	Program kickoff & mobilization	Contract executed; PMO stood up; governance & reporting cadence agreed	☒☒
G1	1	URCA spectrum application filed	VHF (148–174 MHz) license application submitted; frequency plan drafted	☒☒
G2	2	Architecture & site survey baseline	Master Architecture Plan signed; all 15 sites surveyed; link budgets validated	☒☒
G3	3	Pilot hardware procured	Long-lead items (Jetson, MTR3000, Iridium 9603) ordered; supply risk mitigated	
G4	4	First pilot node bench-accepted	One node passes lab acceptance (sensors, 3 radios, power, OTA A/B)	
G5	6	Phase 1 complete — Pilot mesh live 🔄	4 pilots (MYNN, MYGF, MYAM, MYEG) on VHF + Iridium; heartbeat/observation flowing; satellite fallback verified	☒☒
G6	8	Edge AI v1 on pilots	Microclimate + anomaly models running at edge within latency budget	
G7	10	Consensus & autonomous-mode proven	Multi-station consensus vote + hub-disconnect drill pass on pilot cluster	☒☒
G8	12	Phase 2 complete — Intelligence validated	Forecast-verification scoreboard live; MOS bias-correction improving accuracy; autonomy drill signed off	☒☒
G9	14	Southern remote stations live 🔄	MYIG, MYMM, MYRD, MYLD, MYES commissioned satellite-first ; SE edge has verified lifeline before June 1	☒☒🔄
G10	16	Central + northern stations live	Remaining stations (MYBC, MYCB, MYEM, MYER, MYAT, MYBS) joined; full 15-node mesh forming	

Gate	Month	Milestone	Exit criteria (go/no-go)	Critical
G11	18	Phase 3 complete — Full network operational	All 15 nodes commissioned to acceptance spec; emergency VHF + SMS broadcast commissioned; topology resilient to single-node loss	☒☒
G12	21	Advanced capabilities online	Propagation (station-to-station) forecasting; network-wide self-healing & store-and-forward at scale	
G13	23	Operations handover & training complete	Watch officers trained; runbooks delivered; 30-day stability soak passed	
G14	24	Program close & national acceptance	Final acceptance by Ministry / customer; warranty + support contract active; as-built docs delivered	☒☒

4.2 Capability & Integration Gates

These run in parallel with the core gates and inherit the same go/no-go discipline.

Gate	Month	Capability	Exit criteria (go/no-go)	Critical
STR-1	2	Cat 4/5 structural design certified	Tower/mast/enclosure design certified to TIA-222-H / ASCE 7 Risk Cat IV (≥180 mph) + ASTM E1996 missile; per-site surge elevation set; lightning/grounding (IEC 62305) + C5-M corrosion design included	☒☒
GOV-1	4	Data Governance & Sovereignty Policy approved	Policy signed (BACSWN-GOV-2026-005): data classification, residency, retention, sharing rules — required before any inter-agency/external integration	☒☒
ENV-1	2	Site land rights & environmental screening	Land leases/access secured for pilot + designated radar sites; environmental screening done; protected-area constraints identified	☒☒
ENV-2	8	Permits & clearances complete	Environmental impact assessment approved; building + civil-aviation siting permits issued for all station and radar-tower sites	☒☒
VV-1	3	Master V&V plan + ICD register baselined	V&V plan signed (BACSWN-VV-2026-006); requirements-traceability matrix established; all internal/external ICDs registered under change control	☒☒
RAD-1	2	Raytheon radar procurement initiated	Radar siting/coverage analysis done; designated radar-tower sites fixed; Raytheon contract + US export/ITAR licensing filed; long-lead order placed	☒☒
RAD-2	10	Radar-tower civil works complete	Heavy radar-tower foundations/structures built at designated sites (Cat 4/5-rated); power + shelter ready for radar install	
RAD-3	15	First Raytheon radar installed & accepted	First primary radar installed, calibrated, and feeding the surveillance picture; vendor acceptance signed	☒☒
RAD-4	18	Radar coverage operational	Primary radar operational across designated coverage sites; enables military / non-cooperative fusion (SUR-2) and FIR-wide picture	☒☒
SEC-1	3	Crypto root-of-trust baseline	National PKI stood up; secure element/TPM provisioning + measured secure boot proven on pilot node	☒☒
REV-1	6	Overflight/landing-fee metering MVP	Surveillance-driven fee-accounting pipeline live (count → bill basis); reconciles with civil-aviation authority	☒☒
SUR-1	8	Commercial surveillance live	ADS-B/Mode-S airspace picture over the Bahamas FIR; feeds safety + fee + CORSIA	
SEC-2	9	SecOps & first pen-test	Incident response runbook live; RBAC/audit; independent penetration test passed	☒☒
EMG-1	12	CAP/EDXL alerting gateway live	Standards-based alerts ingested by a first agency CAD/dispatch (NEMA)	
SUR-2	15	Military/non-cooperative fusion	Raytheon primary radar (RAD-3) + secondary radar + partner data-share fused into the track picture (Mode 5 / non-ADS-B coverage)	☒☒
EMG-2	16	COP + shelter/evacuation integration	Common Operating Picture shared with NEMA, Police, Fire/Rescue, RBDF; shelter activation/occupancy + surge-aware evacuation routing live	
IOT-1	14	IoT sensor pilot	LoRaWAN pilot cluster reporting via station gateways into the COP	
EMG-3	18	Inter-agency activation drill	Full multi-agency hurricane-activation exercise passes end-to-end	☒☒
DRN-1	20	UAS capability + deconfliction	Drone survey/recon flown with BACSWN live-traffic deconfliction; telemetry into COP; under BCAA UAS rules	
IOT-2	22	IoT network rollout	Priority islands densified; edge-AI fusing IoT into surge/flood nowcasts	
INT-1	22	International exchange live	ICAO OPMET/WAFS/AFTN exchange operating; U.S. (NOAA/NWS/NHC, FAA Miami Oceanic, NORAD) integration scoped/initiated — may extend past M24	

Gate	Month	Capability	Exit criteria (go/no-go)	Critical
DRN-2	23	Drone operations + SAR drill	UAS post-storm + SAR drill with RBDF; comms-relay restore demonstrated	
DR-1	21	Hub disaster-recovery failover drill	Secondary/off-island standby hub + replication proven; COP failover within RTO, near-zero RPO; data restore verified	☒
OM-1	22	O&M / sustainment readiness	NOC, 3 regional service hubs, spares pool, island field-logistics, predictive maintenance & SLAs operational before handover (BACSWN-OM-2026-004)	☒
TRN-1	10	Training programme established	Role-based curricula + competency framework approved; academic partnerships (UB, BTVI, Dept. of Met, BCAA) signed; train-the-trainer cohort under way	
TRN-2	23	Operational certification complete	Bahamian watch officers, NOC operators, and field/radar/drone technicians certified and operationally ready; localization targets on track	☒
VV-2	23	Operational validation complete	Live hurricane-season exercise + full inter-agency drill + hub failover passed; requirements-traceability matrix fully closed	☒
SEC-3	24	Full security acceptance	Network-wide crypto, signed-OTA, tamper-zeroization verified; final pen-test + remediation closed	☒

5. Phase Detail & Sequenced Activities

Phase 1 — Foundation (M1–M6)

- WS-B: File URCA application (**M1, critical path**); open BCAA/ICAO coordination.
- WS-C/D: Finalize link budgets; procure long-lead hardware; build & bench-accept first node.
- WS-E: Mesh protocol MVP (HEARTBEAT, OBSERVATION, routing, store-and-forward) on VHF backbone.
- WS-G: Civil/power at the 4 pilot sites (MYNN, MYGF, MYAM, MYEG); establish Iridium SBD fallback.
- WS-G/J: Cat 4/5 structural design certification (STR-1, M2); national PKI + secure-boot baseline (SEC-1, M3).
- WS-L: Stand up commercial ADS-B surveillance and the **overflight/landing-fee metering MVP (REV-1, M6)** — built early so fee accounting is live as soon as the airspace is monitored.
- WS-P: **Raytheon primary-radar program kickoff** — radar siting/coverage analysis; designated radar-tower sites fixed; Raytheon contract + **US export/ITAR licensing filed**; long-lead radar order placed (RAD-1, M2).
- WS-G: structural design includes **lightning/grounding (IEC 62305) + C5-M corrosion** protection (within STR-1, M2).
- WS-A/J: **Data Governance & Sovereignty Policy** drafted and approved before any external integration (GOV-1, M4).
- WS-Q: Secure **land rights/leases** and complete **environmental screening** for pilot + radar sites; identify protected-area constraints (ENV-1, M2) — this **gates all civil works**.
- WS-H: Baseline the **Master V&V plan + ICD register** and establish the requirements-traceability matrix before integration scales (VV-1, M3).
- Gate G5 (M6)**: pilot mesh live on VHF + satellite. 🌀 *Must clear before hurricane season.*

Phase 2 — Intelligence (M7–M12)

- WS-F: Deploy edge AI (microclimate prediction, anomaly detection) to pilots; stand up forecast verification + MOS bias correction; build ERA5 climate normals.
- WS-E: Consensus voting, propagation message handoff, OTA model/firmware pipeline.
- WS-H: Consensus + autonomous-mode (hub-disconnect) drills on the pilot cluster.
- WS-L: Commercial surveillance operational (SUR-1, M8) feeding safety + fee + CORSIA.
- WS-J: SecOps + incident response live; first independent penetration test (SEC-2, M9).
- WS-K: CAP/EDXL alerting gateway live with NEMA (EMG-1, M12).
- WS-R: **Training programme established** — role-based curricula, academic partnerships (UB/BTVI/Dept. of Met/BCAA), and train-the-trainer cohort begun, with pilot-cluster hands-on (TRN-1, M10).
- WS-Q: **Environmental impact assessment approved**; building + civil-aviation siting permits issued for all station and radar-tower sites (ENV-2, M8).
- WS-F: Integrate the **Tomorrow.io** commercial weather + space-based radar feed (API key already wired) — supplements AWC/NWS/OpenMeteo and adds space-based precipitation radar over open ocean where ground radar is sparse.
- WS-P: Radar-tower civil works underway at designated sites (heavy Cat 4/5-rated structures, power, shelter) toward RAD-2 (M10).
- Gate G8 (M12)**: intelligence validated; autonomy proven.

Phase 3 — Full Network (M13–M18)

- WS-D/G: Roll out remaining 11 stations **south-to-north**, satellite-first for remote sites.
- WS-C: Commission full VHF partial-mesh + Iridium logical-mesh overlay; harden Chub Cay relay.
- WS-E/I: Commission emergency VHF voice + SMS gateway broadcast; begin operator onboarding.
- WS-M: IoT sensor pilot via station LoRaWAN gateways (IOT-1, M14).
- WS-L: Fuse military/non-cooperative radar + data-share into the track picture (SUR-2, M15).

- WS-K: Common Operating Picture + shelter/evacuation integration with NEMA, Police, Fire/Rescue, RBDF (EMG-2, M16); full inter-agency activation drill (EMG-3, M18).
- WS-P: Install, calibrate & accept first Raytheon radar (RAD-3, M15); bring primary-radar coverage operational across designated towers (RAD-4, M18) — unlocking military/non-cooperative fusion (SUR-2).
- **Gate G9 (M14)** and **G11 (M18)**: remote south protected pre-season; full 15-node network operational.

Phase 4 — Advanced Capabilities (M19–M24)

- WS-F: Network-wide propagation forecasting and emergent multi-station hazard detection.
- WS-E: Self-healing at scale; performance hardening; close test-coverage gaps to production bar.
- WS-I: Watch-officer training, runbooks, 30-day soak, knowledge transfer.
- WS-N: Drone/UAS capability with live-traffic deconfliction (DRN-1, M20) and post-storm + SAR operations (DRN-2, M23).
- WS-M: IoT network rollout to priority islands (IOT-2, M22).
- WS-O: ICAO OPMET/WAFS exchange live; U.S./NORAD interoperability initiated (INT-1, M22) — may extend past M24.
- WS-J: Full security acceptance — final pen-test + remediation closed (SEC-3, M24).
- WS-E: **Hub disaster-recovery failover drill** — secondary/off-island hub, replication, RTO/RPO verified (DR-1, M21).
- WS-I: **O&M / Day-2 sustainment readiness** — NOC, regional service hubs, spares, island field-logistics, predictive maintenance & SLAs stood up before handover (OM-1, M22).
- WS-H: **Operational validation** — live hurricane-season exercise + full inter-agency drill + hub failover; requirements-traceability matrix fully closed (VV-2, M23).
- WS-R: **Operational certification** — Bahamian watch officers, NOC, and field/radar/drone technicians certified, drilled, and operationally ready ahead of handover (TRN-2, M23).
- **Gate G14 (M24)**: national acceptance, warranty & support active, program close.

6. Critical Path & Key Dependencies

M0 Kickoff → G1 URCA filing → G2 Arch/site baseline → G5 Pilot mesh (M6,)
↳ G7 Autonomy proven → G8 Intelligence (M12) → G9 Remote-south live (M14,)
↳ G11 Full network (M18) → G14 National acceptance (M24)

- **URCA spectrum (G1)**: longest regulatory lead; any slip cascades to every VHF milestone. Filed M1, target approval by M9.
- **Hurricane windows**: G5 (M6) and G9 (M14) are pinned before June 1. South-to-north sequencing ensures exposed/satellite-critical stations are protected first (risk R01).
- **Supply chain**: long-lead edge hardware ordered at G3; second-source qualified (Raspberry Pi 5
 - Coral, AMD/Xilinx) to de-risk Jetson availability (risk R05).
- **Revenue-enabling (REV-1, M6)**: under the BOO / overflight-fee model, fee metering is on the *value* critical path — prioritized in Phase 1 so billing begins as soon as traffic is monitored.
- **Military surveillance (SUR-2)**: depends on a U.S./defence data-sharing agreement (risk R07); commercial ADS-B (SUR-1) proceeds first and primary radar provides fallback.
- **Security first (SEC-1 → SEC-2 → SEC-3)**: the crypto root-of-trust must precede any external or inter-agency integration, so security gates lead the integration workstreams.
- **Off the URCA path**: the IoT layer (WS-M) uses license-exempt ISM spectrum and does not wait on VHF licensing.
- **Environmental & land (ENV-1/2)**: land rights and permits **gate all civil works** (WS-G) and radar-tower construction (WS-P); started at M0 so site access is never the blocker. Protected-area constraints on the cays can force re-siting, so screening is done before hardware is committed.
- **Raytheon radar (WS-P)**: alongside URCA, this is a top long-lead critical item. **US export/ITAR licensing** and radar manufacturing lead times start at M1–M2; heavy radar-tower civil works (RAD-2) then gate installation (RAD-3, M15). The military/non-cooperative surveillance picture (SUR-2) and full FIR coverage depend on radar being operational (RAD-4, M18) — so any radar slip cascades into the surveillance and (partially) the fee/coverage value case.

7. Top Program Risks (schedule-impacting)

The **full SWOT & consolidated risk register** is maintained separately in **BACSWN-RISK-2026-008** (SWOT & Risk Analysis). The table below is the schedule-impacting subset; the consolidated document is authoritative for scoring, residuals, and governance.

ID	Risk	L×I	Mitigation	Owner
R01	Hurricane strikes partially-deployed network	4×5 CRITICAL	South-to-north sequencing; satellite fallback live before June 1	WS-A/C
R03	VHF spectrum delay (URCA)	3×4 HIGH	File M1; maintain satellite/cellular interim path	WS-B

ID	Risk	LxI	Mitigation	Owner
R04	Edge AI underperformance	3x3 MEDIUM	Hub-based models as fallback; phased validation per gate	WS-F
R05	Edge hardware supply disruption	— MEDIUM	Pre-order at G3; qualified second sources	WS-D
R06	Air-traffic volume downturn cuts overflight/landing-fee revenue	3x4 HIGH	Diversified airline base over a busy oceanic FIR; minimum-traffic floor in the concession; opex flexibility	WS-A/L
R07	U.S./military data-sharing agreement delayed (military track fusion)	3x3 MEDIUM	Commercial ADS-B first; pursue MOU early; primary-radar fallback	WS-L/O
R08	Drone/UAS regulatory approval (BCAA)	2x3 MEDIUM	Early BCAA engagement; leverage own air picture for deconfliction	WS-N
R10	Raytheon radar long-lead, US export/ITAR delay, or install complexity	4x4 CRITICAL	File ITAR/export + place order at M1; radar-tower civil works in parallel; phased install; commercial ADS-B + Tomorrow.io space-radar as interim coverage	WS-P/B
R11	Tomorrow.io feed dependency (commercial API availability / cost)	2x2 LOW	Treat as supplementary, not sole-source; AWC/NWS/OpenMeteo + own sensors remain authoritative	WS-F
R12	Environmental clearance / land rights / permitting delay (gates civil works)	3x4 HIGH	Start at M0; engage Ministry + environment authority early; prioritize pilot + remote/radar sites; sequence civil works behind cleared sites; hold re-siting options	WS-Q

8. Governance & Reporting Cadence

- **Weekly:** PMO status to the owner/CEO (RAG, milestone burn-up, risk deltas).
- **Per gate:** formal go/no-go review with documented exit-criteria sign-off.
- **Monthly:** owner/CEO review — milestone progress and forward-look.
- **Change control:** any scope/cost/schedule change >5% routed through WS-A change board.

Companion documents: *BACSWN Mesh Architecture Master Plan (BACSWN-MARCH-2026-002)*; *Investor Operations & Maintenance Plan (BACSWN-OM-2026-004)*; *Data Governance & Sovereignty Policy (BACSWN-GOV-2026-005)*; *Master V&V and ICD Plan (BACSWN-VV-2026-006)*; *Training & Capacity-Building Plan (BACSWN-TRN-2026-007)*; *SWOT & Risk Analysis (BACSWN-RISK-2026-008)*; *Board Runbook (technology highlights)*.